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From Kernels to Standards: How Kokkos and Modern C++ Future-Proof Critical Simulation Codes

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U.S. DEPARTMENT
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FRONTIER

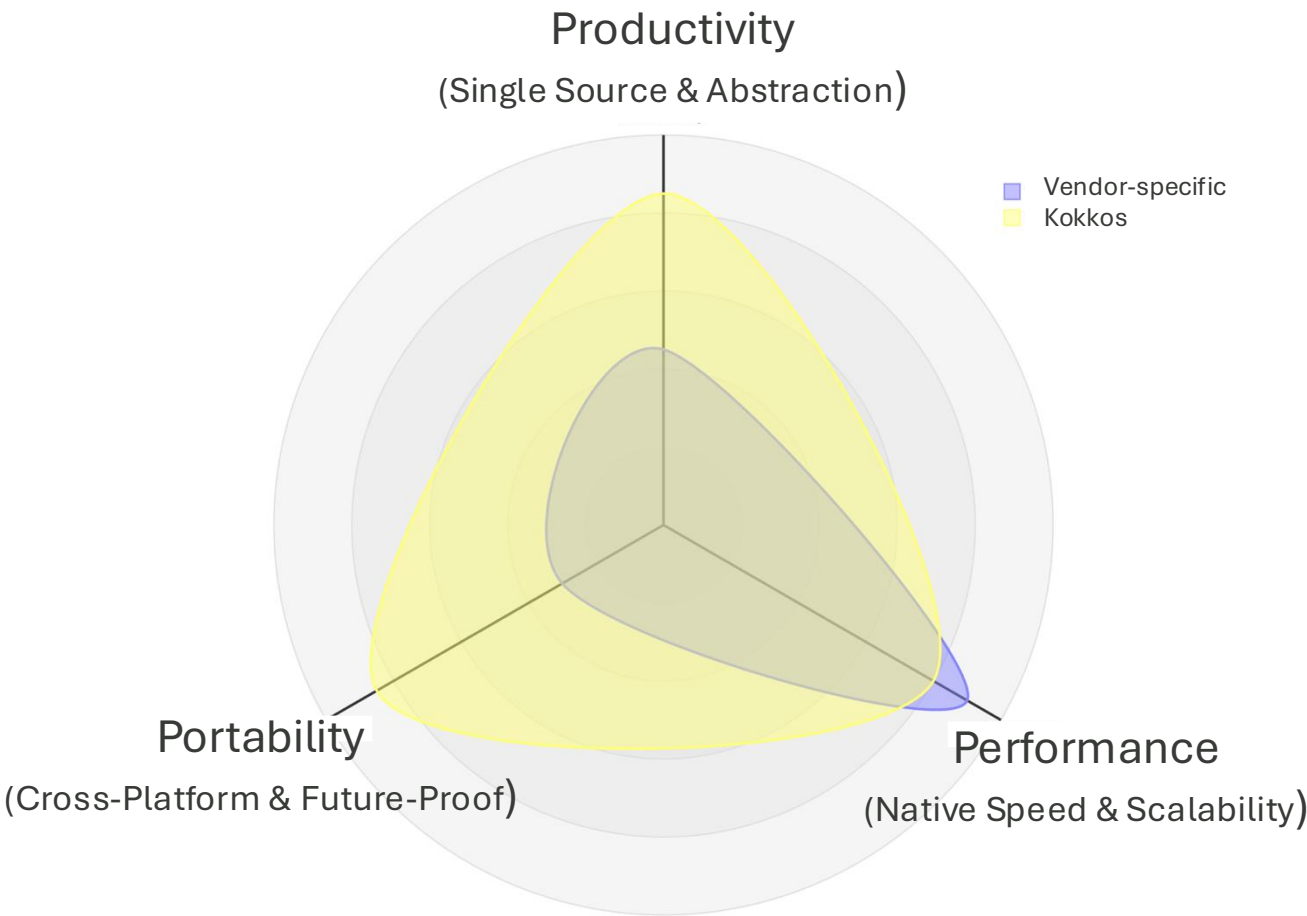
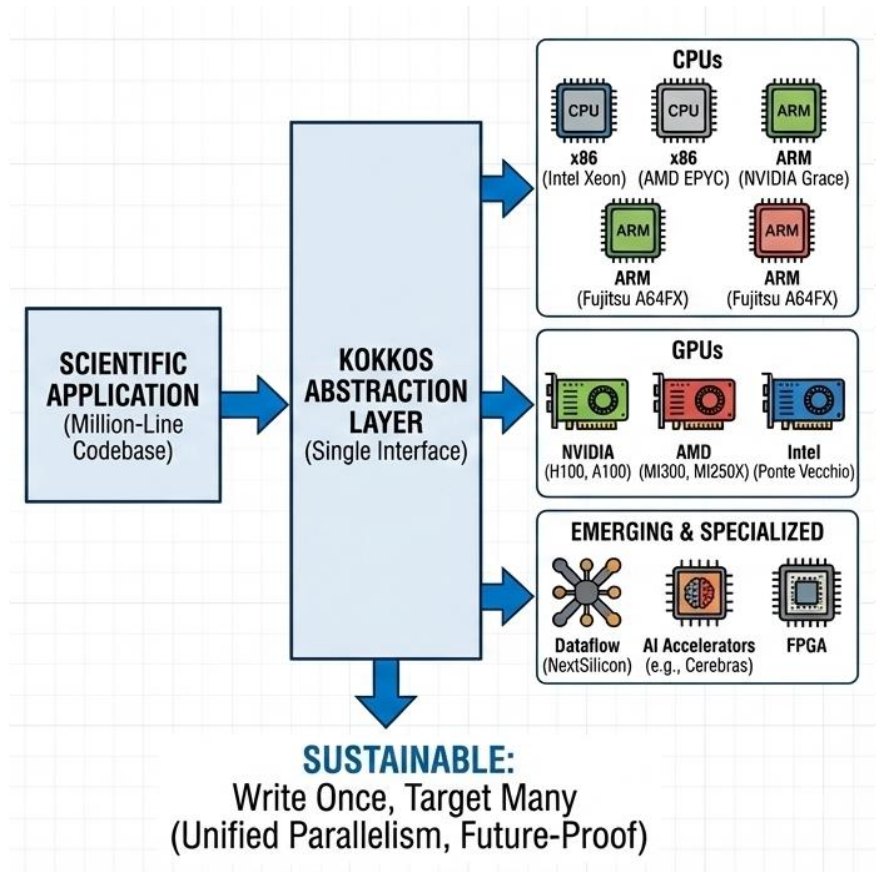


Acknowledgments

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The Exascale Era Challenge



The “P3” Challenge

Three Pillars to Tackle the Exascale Challenge



Kokkos



Standard C++



HPSF

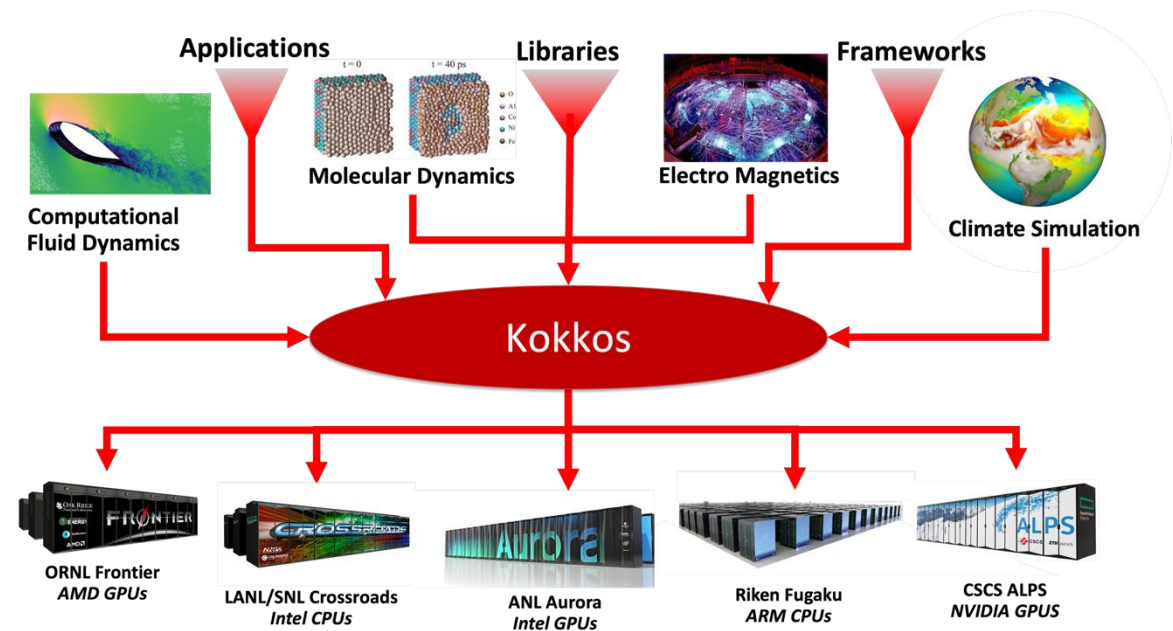
HPSF

The "Kernels" – Kokkos and Performance Portability



kokkos

The Kokkos Programming Model



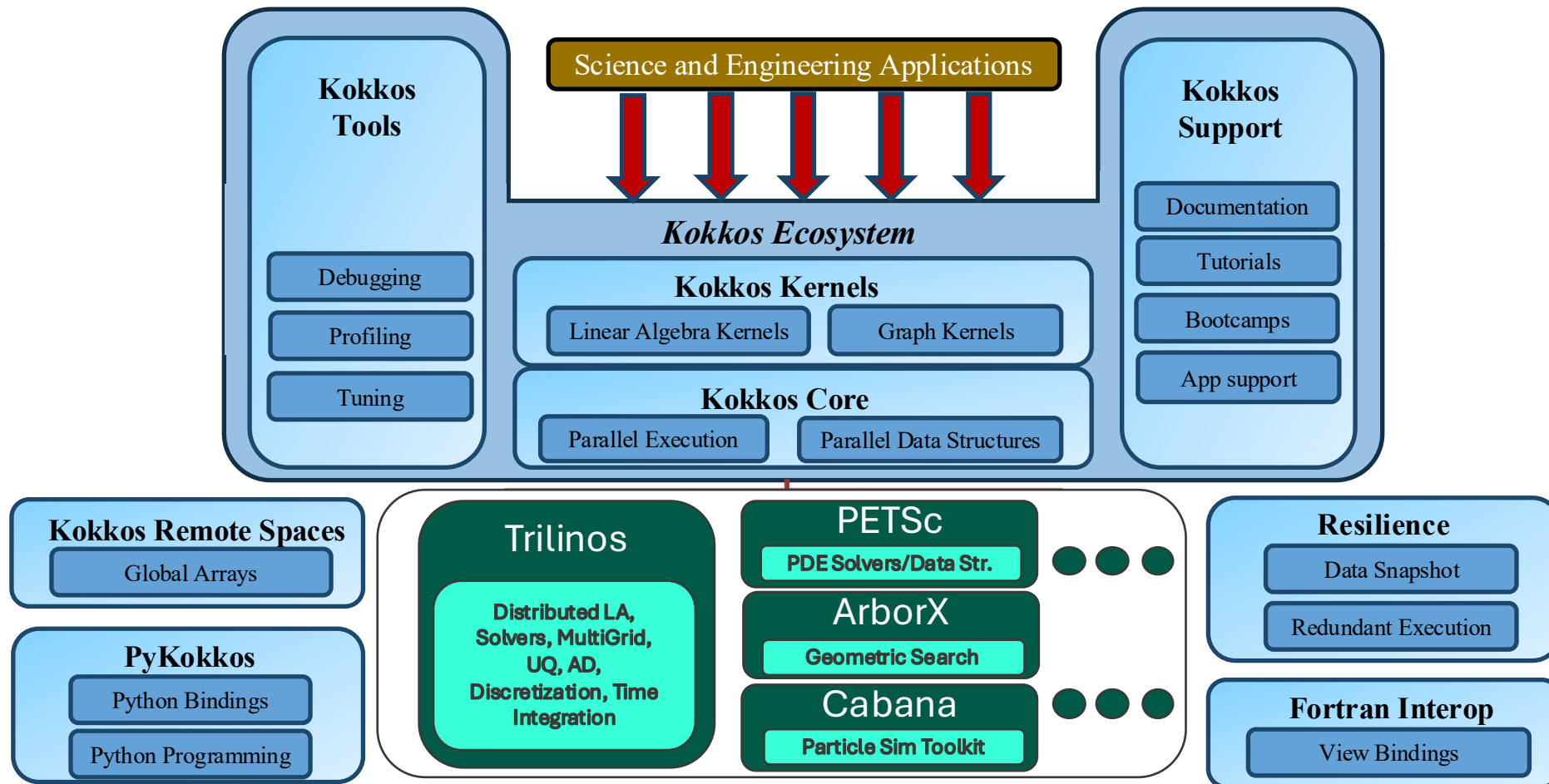
What is Kokkos?

- A C++ Programming Model for Performance Portability
 - Implemented as a template library on top of CUDA, OpenMP, .
 - Aims to be descriptive not prescriptive
 - Aligns with developments in the C++ standard
 - Replaces usage of CUDA, OpenMP, HIP, etc.
 - Use each vendors toolchain!
- Expanding solution for common needs of modern science/engineering codes
 - Math libraries based on Kokkos
 - Tools which enable insight into Kokkos

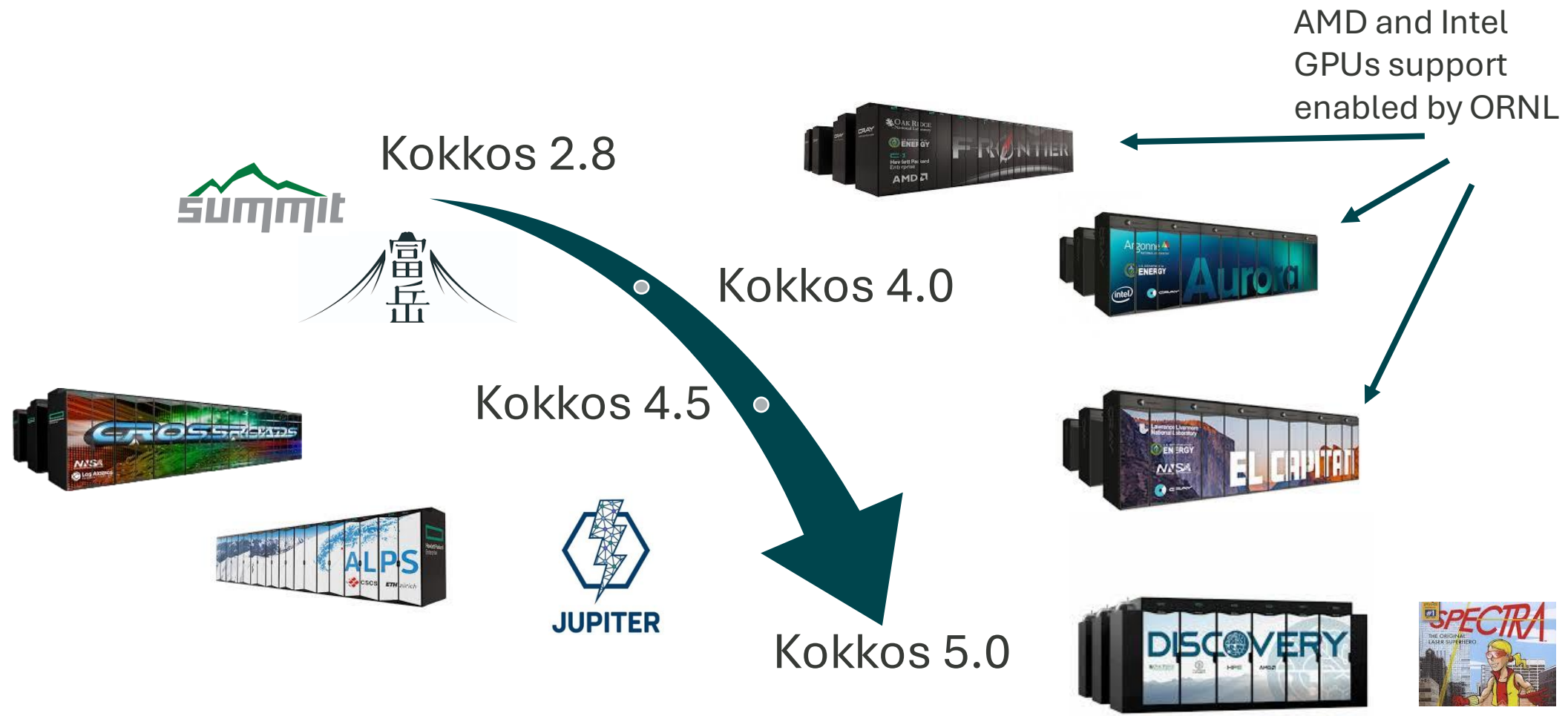
Kokkos is NOT just for GPUs!



More Than Just a Programming Model: A Comprehensive Ecosystem



Powering the DOE Exascale Trilogy



Kokkos at ORNL: Driving Mission-Critical Science

Enabling Libraries

- Geometric Search (ArborX)
- Particle-based Simulations (Cabana)
- Particle-In-Cell (Picasso)
- Mesh Generation (Halyard)
- Parallel I/O (ADIOS)
- Performance Portability and Fortran Code Porting (YAKL)

Core Developer Team

5 Dedicated Dual-Role Experts
(30%-60% Effort on Kokkos)

#1 contributor rank (last 4y)

Lead CI and Build & Packaging
WGs

OLCF support

Post-ECP software stewardship
funding

Powering Applications

- Fusion (VERTEX)
- Additive Manufacturing (Toucan, ExaCA, PicassoMPM)
- Fracture Mechanics (CabanaPD)
- Enrichment (Spinnaker)
- Combustion (Quilt, Grit)
- Quantum Materials (DMRG++)

We don't just develop Kokkos; we ensure it works for ORNL's mission-critical science

Impact Case Study: ArborX



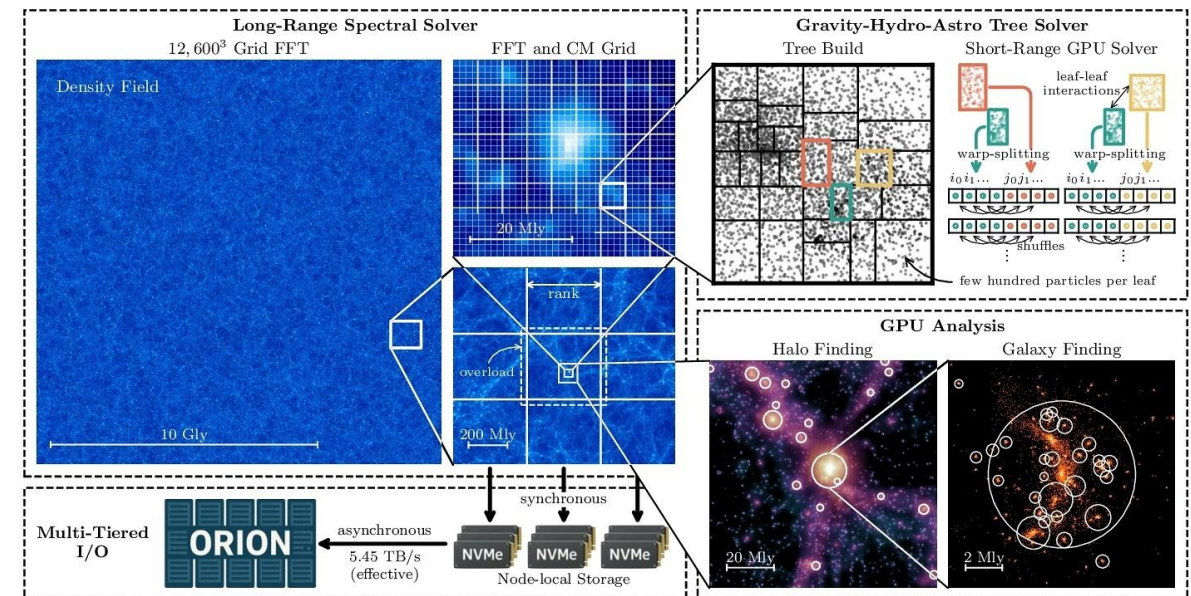
Spatial Indexing and Clustering Library

- Provides data structures to accelerate searching for objects in space
- Includes support for k-NN, ray tracing, collision detection, particle neighbor lists, interpolation, etc.
- Co-designed experimental feature to lower the GPU occupancy

Featured in 2025 Gordon Bell Prize Finalist

“Cosmological Hydrodynamics at Exascale: A Trillion-Particle Leap in Capability”

- Largest ever cosmological hydrodynamics full-sky simulation (> 10x state-of-art)
- Required 9000 nodes for one week on Frontier
- Achieved 513 PFLOPs peak performance
- Produced 100 PB of cosmological data



<https://www.olcf.ornl.gov/2025/11/14/largest-ever-universe-simulation-up-for-supercomputings-highest-prize/>
<https://doi.org/10.1145/3712285.3771786>

Building a Community

Developer Team

Primary
Teams



Sandia
National
Laboratories



Support
Efforts



Helped us get there:



EXASCALE
COMPUTING
PROJECT

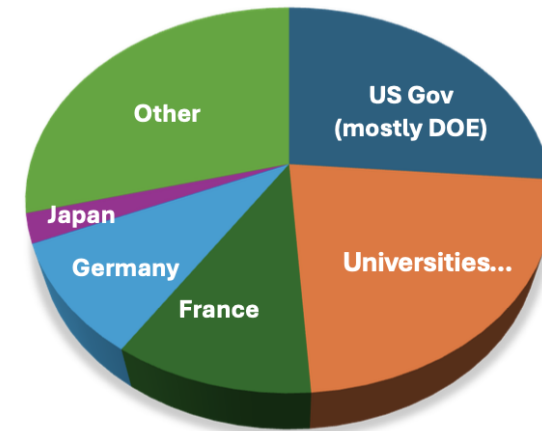
Supports us in our journey to the next level:



Users Community

- Thousands of users (2500 registered on Slack)
- Hundreds of institutions and projects

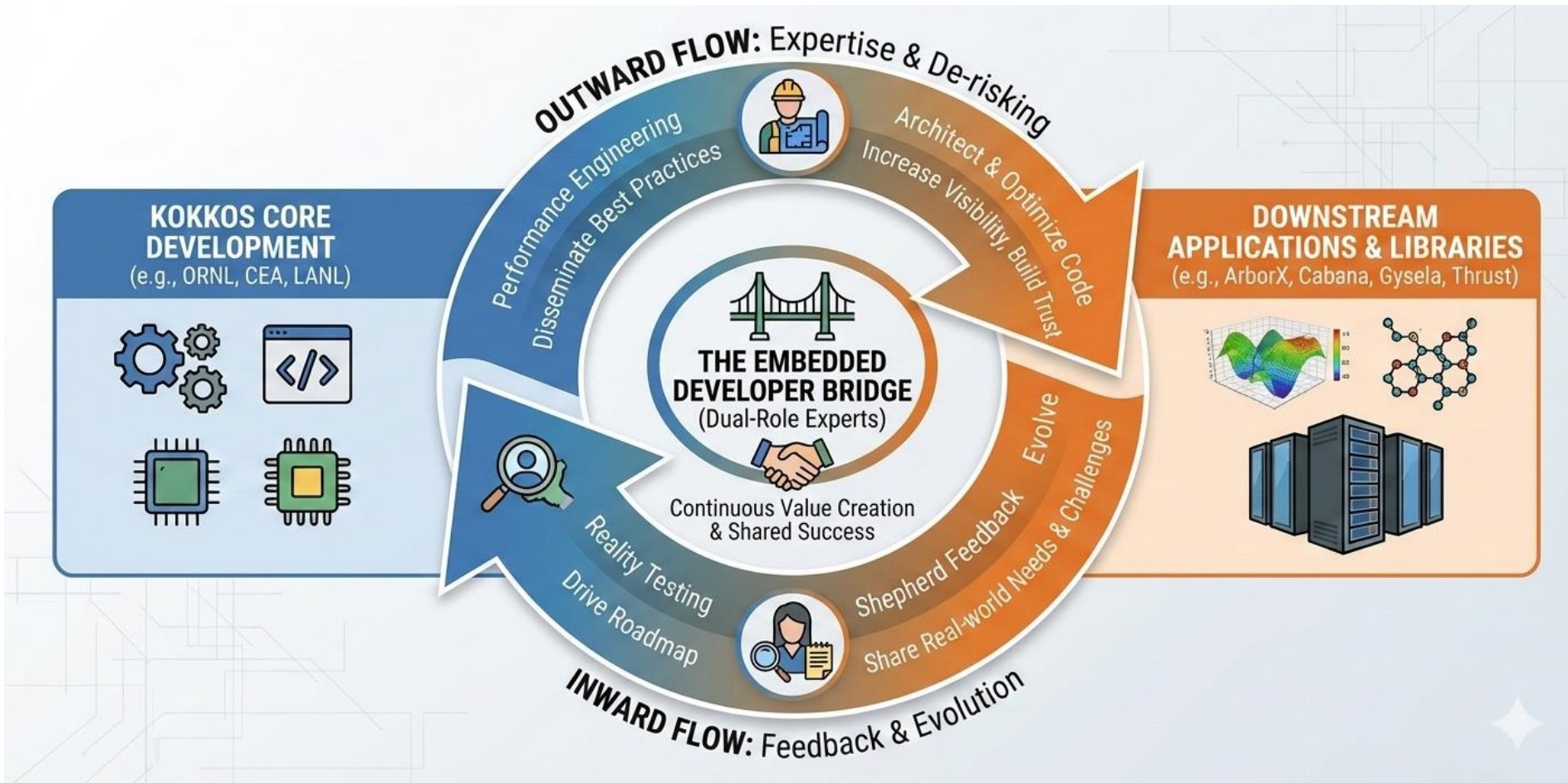
Slack User Origin



HPSF

The Embedded Developer Model

Maximizing Kokkos Return On Investment



Hardening the Ecosystem: Production-Grade Reliability

Structural Governance: Investing in the Future

- **Dedicated Working Groups:** Established “CI” and “Build & Packaging” WGs to move beyond "ad-hoc" maintenance and toward institutionalized quality standards.
- **Reinvestment Rule:** Dedicating 10% of core effort to managing technical debt and paying down the principal
- **Long-term Stability:** Ensuring the core remains agile to be ready for what’s coming next

Rigorous Testing: Integration at Scale

- **Ecosystem Integration:** We don't test in a vacuum; we run continuous integration with downstream partners (e.g. ArborX ,Trilinos, PETSc, LAMMPS) to catch regressions before they impact the science.
- **Universal Coverage:** Continuous testing across the full landscape:
 - On-Prem: Sandia and ORNL internal resources
 - Leadership Facilities: OLCF, ALCF, and NERSC systems

Advanced Tooling: Detecting the Invisible

- **Static Analysis:** Developing Clang-Tidy extensions for invalid code detection and automated migration.
- **Dynamic Safety:** Kokkos Tools recently resolved critical "View of Views" hangs in production code.
- **The Next Frontier:** Increasing investment in tooling to move from bug detection to automated prevention.

The "Standards" – Bridging to ISO C++



The Strategy of Language Evolution

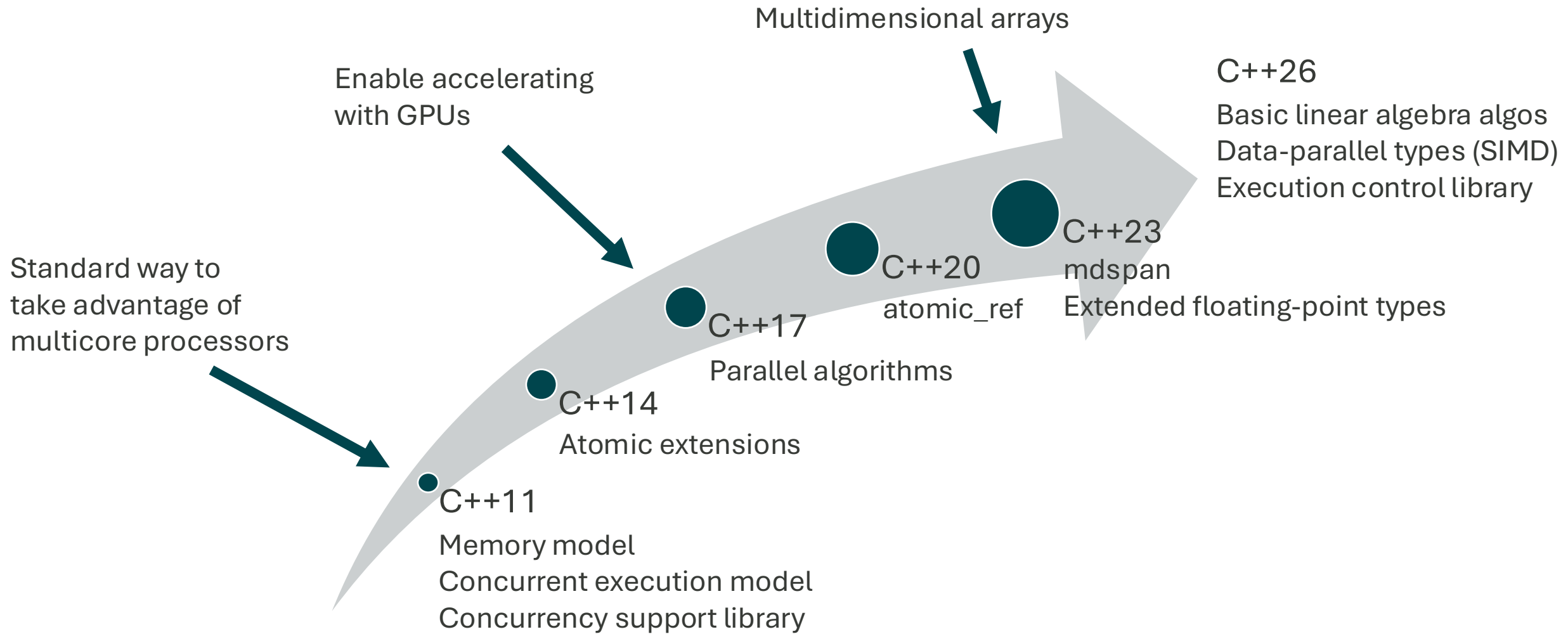
The Problem with Libraries: Even the best libraries represent a dependency. Long-lived codes (30+ years) require foundational stability.

Benefits of Standardization:

- **Vendor Neutrality:** Compilers (GCC, Clang, MSVC, Intel, NVIDIA) must support the standard
- **Interoperability:** Different libraries (e.g., solvers, I/O, viz) can communicate via standard types rather than custom wrappers.
- **Reduced Technical Debt:** Lowering the barrier to entry for new developers by using standard C++ features taught in universities.

Role as DOE lab representative: Ensuring that the unique constraints of HPC are core to C++'s evolution

HPC Features by Standards



Success Story: mdspan (Multi-dimensional arrays in C++23)

```
Kokkos::View<double**> A("A", M, N);  
Kokkos::View<double[4][4], Kokkos::LayoutLeft> B("B");
```

```
std::mdspan A(ptr, M, N);  
std::mdspan<double, std::extents<int, 4, 4>, std::layout_left> B(ptr);
```

```
template <  
  class DataType  
  [, class LayoutType]  
  [, class MemorySpace]  
  [, class MemoryTraits]>  
class Kokkos::View;
```

```
template<  
  class T,  
  class Extents,  
  class LayoutPolicy = std::layout_right,  
  class AccessorPolicy = std::default_accessor<T>  
> class mdspan; (since C++23)
```

Challenge: C++ lacked a native way to represent multidimensional arrays

*P0009 **mdspan**: A Non-Owning Multidimensional Array reference*

Impact: Kokkos-inspired feature now allows scientific codes to share data across libraries seamlessly

kokkos/mdspan backport to C++17 deployed in NVHPC
Contributed an implementation to libc++ (LLVM)
Kokkos::View refactored in terms of mdspan in release 5.0

The Next Frontier: Standard Linear Algebra (C++26)

```
dgemv('N', M, N, 1., A, 1, x, 1, 0., y, 1); // 11 parameters      BLAS
KokkosKernels::gemv('N', 1., A, x, 0., y);                      KokkosKernels
std::matrix_vector_product(A, x, y);                            Standard C++
```

```
template<
    [class ExecutionHandle,]
    class InMat,
    class InVec,
    class OutVec>
void KokkosKernels::gemv (
    [const ExecutionHandle& exec,]
    const char trans[],
    typename InMat::const_value_type& alpha,
    InMat A,
    OutMat x,
    typename OutVec::const_value_type& beta,
    OutVec y);
```

```
template<
    [class ExecutionPolicy,]
    InMatrix InMat,
    InVector InVec,
    OutVector OutVec
> void matrix_vector_product( [ExecutionPolicy&& exec,]
                               InMat A,
                               InVec x,
                               OutVec y ); (since C++26)
```

Scientific computing relies on BLAS/LAPACK, but calling these often requires archaic Fortran interfaces or vendor-specific wrappers

P1673: A C++ Standard Library Interface for Parallel Linear Algebra

What it Delivers:

- A modern, type-safe interface for vector and matrix operations
- Support for execution policies (running on GPUs/Multi-core via `std::execution`) and mixed precision
- A common language for developers to express mathematical intent that vendors can optimize under the hood

Strategic Value: Future-proofs computationally expensive parts of simulation codes by making linear algebra a first-class citizen of the C++ language

Leading the HPC Voice in the C++ Community

Serving as "HPC Filter" in WG21: Review incoming proposals to ensure they don't inadvertently harm performance or ignore the needs of large-scale distributed systems

Community Liaison: Bridging the gap between lab scientists and the C++ community at large

- Scientific Computing Track at CppCon
- Disseminate best practices and share lessons learned from other domains (Embedded, Game dev, Trading, etc.)



Goal: Make C++ the most productive language for AI and post-exascale science without losing the "zero-overhead" performance guarantee

Strategic Future- Proofing – HPSF & Leadership



HPSF

HIGH PERFORMANCE
SOFTWARE FOUNDATION

Beyond the Code: The High Performance Software Foundation

The Vision:

Creating a neutral, vendor-agnostic home under the Linux Foundation to host and sustain the essential software stack for global HPC.

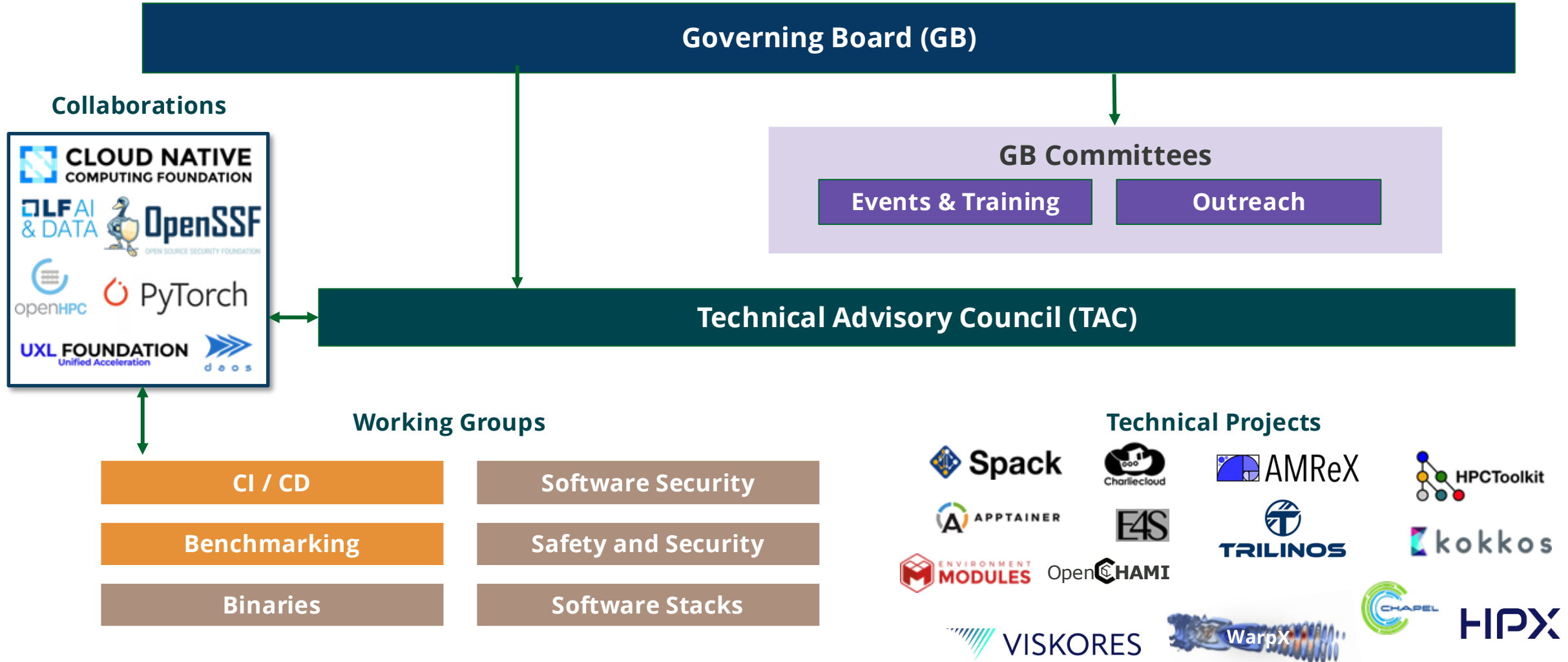
Value to the Labs:

- Ensures that critical tools have a life beyond a single funding cycle.
- Fosters a "neutral ground" for DOE, industry (AMD, NVIDIA, Intel), and international partners to collaborate.

Leadership Role:

- **Co-Founder:** Collaborated with industry and lab partners to define the foundation's core mission and governance.
- **Governing Board Member:** Driving strategic decisions on budget, policy, and cross-foundation resource allocation.
- **Technical Advisory Council (TAC):** Providing technical oversight and ensuring interoperability across flagship projects like Kokkos, Spack, Trilinos, Viskores, HPCToolkit, etc.

HPSF Charter



HPSF Members

Premier



General



Associate



BERKELEY LAB



UNIVERSITY OF OREGON



What's in it for Projects?



HPSF

Vendor-Neutral Home & Stability

Ensures project independence and long-term sustainability under the Linux Foundation.

Growth, Branding & Focus

HPSF handles legal, marketing (websites, new logos), and community growth, letting your team focus on the code.

Infrastructure & Automated Testing

Access dedicated hardware and CI/CD Working Group support for robust automated testing and benchmarking.

Collaboration & Events

Host user groups at the annual HPSF Conference and facilitate multi-institutional collaboration and workforce development.

HPSF Impact: Professionalizing the Kokkos Ecosystem

Governance & Operational Maturity

- Formalized project governance and started developing a community playbook.
- Core leadership graduated from the CSCCE Training for Open-Source Leaders (birdaro 2025 program).
- Committed to the OpenSSF Best Practices Program for security and sustainability.

Engineering & Infrastructure

- Secured early access to hardware for pre-release testing, such as NVIDIA Blackwell.
- Implemented LFX Tools and Slack Pro to manage and analyze a community of 2,500+ users.
- Utilizing Sustainable Research Pathways to diversify and grow the developer pipeline.

Transparency & Engagement

- Established regular Release Briefings and recorded technical seminars.
- Institutionalized Kokkos User Group (KUG) meetings and the Tea-Time Seminar Series.

International Strategic Partnerships

Aligning Global Roadmaps: Actively bridging the gap between US, European, and Japanese HPC efforts.

DOE/MEXT (US-Japan): Leading working groups on Programming Languages and Scientific Libraries to ensure our exascale stacks remain compatible and state-of-the-art.

ADAC: Driving international standards for "Portability, Sustainability, and Integration" in large-scale codes.

Strategic Outcome: Reducing redundant development by leveraging global expertise and shared software components.



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EuroHPC
Joint Undertaking

Cultivating the HPC Community

Expanding the C++ Ecosystem:

- Founded and Chaired the Scientific Computing Track at CppCon
- Created a dedicated space for scientists within the world's largest C++ conference to influence the "mainstream" developer community.

Organizational Service:

- 2nd edition of the HPSF Conference in Chicago March 19-20, 2026
- ISO C++ panel to discuss the latest development of the language and tutorials to teach how to leverage standard parallelism on heterogeneous systems (SC, ISC, HPC Asia)

Mentorship & Workforce Development:

- Engaging with application and other software teams
- Active mentorship of post-docs and junior staff
- Hosting student interns and collaborating with faculties
- Ensuring the "Performance Portability" mindset is passed down to the next generation of computational scientists

Conclusion

Code (Kokkos)

High-performance, portable
implementation



Standards (ISO C++)

Language-level stability and
longevity



Governance (HPSF/Consortia)

Organizational sustainability
and community health

Looking Forward:

- Beyond exascale: AI-integrated HPC and specialized hardware.
- Our commitment: Continuing to bridge the gap between low-level hardware kernels and the high-level needs of the scientific mission.

Thank you – Questions?

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